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Artificial intelligence with metasurfaces: from intelligent design to intelligent computing

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Abstract

The rapid development of artificial intelligence (AI) has revolutionized traditional design and optimization approaches for metamaterials. By leveraging AI techniques, the time required for the metamaterial design process is significantly reduced, thus improving the overall efficiency. On the other hand, the advancement of AI has placed extremely high demands on computational power. In exploring high-speed and energy- efficient next-generation computing hardware, metasurface-based intelligent computing technologies such as diffractive neural networks have attracted widespread attention. In this work, we focus on two main aspects: intelligent design of metasurfaces and intelligent computing based on metasurfaces. In the field of intelligent design, we analyze how deep learning-based inverse design methods overcome the efficiency bottleneck of traditional electromagnetic simulations and enable high-precision and automated generation of subwavelength structures. In intelligent computing, we comprehensively explore the implementation mechanisms of optical diffractive neural networks and microwave programmable neural networks based on metamaterials. By comparing the application performance of physical neural networks with different architectures in scenarios such as image recognition and wireless communication, we reveal the crucial role of deep integration of the metasurfaces and AI in driving new computing paradigms, providing a theoretical framework and technological roadmap for next-generation intelligent electromagnetic systems.

Introduction

In the past twenty years, metamaterials [1] have been widely studied and utilized due to their unique physical properties and unparalleled ability to control wave behavior. Metamaterials are arrays designed from uniquely shaped scattering units, exhibiting special electromagnetic responses such as negative refractive index media [2], zero refractive index materials [3], and materials with extremely high refractive indices [4]. In recent decades, many devices have emerged based on the properties of metasurfaces, including superlenses [5], invisibility cloaks [6], absorbers [7], vortex beam generators [8], and holography [9]. Traditionally, metasurfaces have been manipulated by varying their dimensions continuously to alter their electromagnetic characteristics. With the advancement of modern information science, discretizing the metasurfaces into binary encodings has become essential. Cui et al. proposed the concepts of digital coding and

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programmable metasurfaces that use the digits "0" and "1" (having 0° and 180° phase responses) to characterize the metasurfaces [10], thus controlling the electromagnetic waves through different coding sequences. This establishes a link between metamaterial physics and information science technologies. The introduction of information metamaterials has laid an important foundation for informatization and intelligent applications of metamaterials. Based on this concept, numerous practical applications have been proposed, such as wireless communication [11–13], intelligent sensing [14, 15], adaptive control [16, 17], and intelligent computing [18, 19].

Over the past few years, remarkable progress has been made in designing and optimizing metasurfaces. With the rapid development of artificial intelligence (AI), AI algorithms have been introduced to revolutionize the design process of metasurface units. By integrating deep learning algorithms with metasurface design, a great number of related works have been implemented for the intelligent design of meta-atoms, which not only reduces the computational demands but also ensures effective design outcomes. In traditional electromagnetic design, the process begins with modeling to construct units or metasurfaces, followed by full-wave simulations using commercial software such as CST Microwave Studio to obtain the corresponding S-parameters, as well as reflection and transmission characteristics within specific frequency ranges. Changes to the structure are typically made based on experience or electromagnetic optimization knowledge, and these modifications and subsequent full-wave simulations constitute the bulk of the work. Employing AI in this process could significantly reduce the time spent on design optimization.

For training artificial intelligence networks, the objective is to enable the network to understand the actual goals of the task, such as generating unit structures based on S-parameters or classifying existing units based on certain characteristics. Generally, calculating the performance parameters of existing units is known as forward design. Thus, the process of automatically generating unit structures when given specific targets is referred to as inverse design of metamaterials, and has been widely proposed based on AI. In 2017, a software-based automatic-approach, which combines the binary particle swarm optimization (BPSO) with commercial electromagnetic simulation software, was presented for the automatic design of meta-atoms by Zhang et al [20]. In 2019, a deep learning-based neural network method was further proposed for designing target meta-atoms [21]. In the same year, An et al. [22] proposed a specialized deep predictive neural network (PNN) to predict the complete scattering performance of three-dimensional all-dielectric meta-atoms. Besides, Zhu et al. developed a rapid and precise inverse design method for units using transfer learning [23], based on the GoogLeNet-Inception-V3 architecture with an accuracy of 90%. Liu et al. utilized Generative Adversarial Networks (GAN) to implement a method for the inverse design of metasurfaces based on input spectrum [24]. Itzik et al. developed a fully connected neural network designed to predict the spectra associated with complex nanostructures, comprising a Geometric Prediction Network (GPN) and a Spectral Prediction Network (SPN) [25]. The GPN inversely predicts geometric shapes based on their spectra, and the SPN subsequently predicts the corresponding spectra based on the outputs from the GPN. Considering that a single network may not fully capture the complex nuances in practical applications, Jia et al [26]. hence presented a synthetical neural network, inheriting knowledge

from different neural networks for the inverse design of different metasurfaces. Additionally, Xu et al. [27] proposed a metasurface design approach based on transfer learning, which reduces reliance on large-scale data by bridging the gap between different operating frequencies.

Although AI technology is evolving rapidly, the development of AI computing power based on electronic computers is currently facing bottlenecks due to chip manufacturing processes and the von Neumann system. Therefore, finding a new type of computing hardware with high efficiency and high computing power is the key. Recently, the advent of artificial physical neural networks has revolutionized the landscape of electromagnetic wave processing, enabling unprecedented advancements in computational efficiency and functional capabilities [28–39]. In particular, optical neural networks (ONNs) and diffractive neural networks (DNNs) emerged. By leveraging the principles of diffraction and interference, these neural networks manipulate the propagation of electromagnetic waves to perform complex computations. Optical diffractive neural networks use the wave nature of light to encode, transform, and decode information [32, 40–42]. The utilization of optical elements, such as phase and amplitude masks, allows for the modulation of light at various spatial frequencies. These modulated light fields, when passed through a sequence of diffractive layers, facilitate the realization of intricate functions akin to those performed by conventional electronic neural networks. The inherent parallelism and high bandwidth of optical systems render them exceptionally suitable for high-speed data processing applications [43–48]. Compared to the DNN in real space, the DNN at the Fourier plane of an optical system can theoretically address more complex tasks. In addition, using a Fourier-space DNN can further process more complex recognition (computer vision) tasks [49]. To further improve the computing performance of DNNs, Dai and co-researchers reported an optoelectronic fused computing architecture based on the reconfigurable diffractive units [50]. In addition, the optical-electronic hybrid neural network based on a single-layer all-dielectric metasurface has been proposed [51]. Dual-adaptive training allows the neural networks to adapt to substantial systematic errors and maintain their prediction performance [52].

However, the three-dimensional structure of DNNs poses significant challenges for on-chip integration. In addition, misalignment issues during fabrication can degrade network performance. Two main approaches have been proposed to address these issues: new physical architectures and novel training methods. New planar integrated on-chip diffractive neural networks consist of multi-layer metasurfaces, where each diffractive meta-unit is a silicon slot group filled with silicon dioxide [53, 54]. Meanwhile, optical integrated circuit neural networks (ONNs) have attracted much attention owing to their low path loss and easy integration [32, 33, 55–65]. The ONN framework based on Mach–Zehnder Interferometers (MZIs) is mainly constructed through a singular value decomposition algorithm. ONNs can successfully process different optical information, such as speech recognition and handwritten digit classifications, by dynamically controlling temperatures of the titanium nitride (TiN) loaded on the MZI structures. Besides, the all-optical spiking neural network framework, consisting of a wavelength division multiplexer and optical waveguide loaded with phase-change materials, has been demonstrated experimentally that it has self-learning capabilities [33]. For the previous works, the network is mainly used for

predicting information. However, the corresponding training tasks are processed on a computer, which will cost a large amount of time and energy. Recently, Sunil Pai and co-researchers demonstrated experimentally in situ backpropagation for deep learning in photonic neural networks [58, 59].

Similarly, in the microwave domain, the physical neural networks exploit the longer wavelengths and lower frequencies of microwaves to achieve efficient signal processing [66]. The design of microwave DNNs often incorporates the metasurfaces or metamaterials [16], engineered to control the electromagnetic wave propagation [67, 68]. These structures enable the implementation of desired phase and amplitude relationships, allowing for the manipulation of microwave signals in a manner analogous to optical DNNs. The lower attenuation and greater penetration depth of microwaves make them ideal for applications in recognition, imaging, sensing, and communication.

However, DNNs at optical and microwave frequencies do not have programmability due to the lack of adjustable components, resulting in the difficulty of multi-task for the same structure. In order to obtain a richer capacity of DNNs to process information, programmable DNNs at microwave frequency have been reported by using information meta-surface, where some active amplifiers are integrated into the meta-units. This network not only inherits the light-speed processing information of optical DNNs but also owns the capacity of the programmability [19]. In 2023, Cui et al. proposed spoof plasmonic neural networks for microwave processing and detection [18, 69], where electromagnetic waves can be manipulated freely owing to the flexible dispersion behaviors of spoof surface plasmon polaritons [70–74]. Particularly, the nonlinear activation function can be customized via the relationship between the integrated power amplifier and the detectors. More importantly, this network has a flexible structure framework compared to the other physical networks. Programmable DNNs and spoof plasmonic neural networks have been demonstrated to be regarded as decoders in wireless communications. Compared to traditional communication systems, the decoder based on neural networks can replace mixers, multiplexers, and other RF components. More importantly, the decoding speed is no longer limited by the baseband processing clock rate.

Distinct from the prior reviews on similar topics, here we provide a systematic view of recent progress on two principal and interconnected threads: intelligent design of metasurfaces and metasurface-enabled intelligent computing. We begin by introducing the foundational concept of programmable and information and metamaterials, alongside their typical application domains. Subsequently, we detail how the artificial intelligence drives intelligent design process of metamaterials in Chapter 2, covering optimization methods for both individual meta-atoms (Sect. 2.1) and large-scale metasurface arrays (Sect. 2.2), followed by representative applications of the designed intelligent metasurfaces (Sect. 2.3). In Chapter 3, we shift our focus to metasurface-based intelligent computing. We survey the evolution from free-space diffractive neural networks (Sect. 3.1) to chip-integrated photonic networks (Sect. 3.2) and programmable metamaterial computing systems in the microwave regime (Sect. 3.3). A quantitative performance comparison across these architectures is provided in Sect. 3.4. Finally, we synthesize the mutually reinforcing relationship between the intelligent design and computing, and summarize the key challenges, emerging applications, and future research directions that highlight the co-evolution of both fields in the conclusion part.

Intelligent algorithms for metasurface designs and applications

In recent years, artificial intelligence has garnered widespread attention and has been extensively applied to the design and optimization of metasurfaces. In 2019, Zhang et al. proposed a method for automatically designing metasurface units with arbitrary phase responses [20]. Samad and colleagues utilized an adaptive genetic algorithm (AGA) to automatically design units composed of quad-symmetrical binary gold patterns, ensuring specific reflection coefficients [75]. Sajedian et al. introduced the Double Deep Q Network (DDQN) into unit design [76], selecting suitable results from a large dataset by changing materials and unit dimensions, significantly reducing the time required for optimization design.

Intelligent algorithms for meta-atom design and optimization

One of the most crucial aspects of machine learning or deep learning is defining the problem for the computer, essentially enabling it to understand the problem. In the intelligent design of meta-atoms, the first step in the optimization process is to input the meta-atom's geometry into a computer. Directly inputting the CAD model of meta-atoms is highly discouraged due to its complexity and the abundance of redundant structural information. Instead, the recommended approach is to parameterize the meta-atom structure and use these structural parameters as optimization variables. Similarly, it is crucial to define the optimization target using a target function or error function. In general, the forward process involves predicting the target function based on the given structure of meta-atoms. Conversely, the inverse process refers to automatically generating the structure of meta-atoms based on a given design objective [77]. Figure 1 outlines the general paradigm of a deep neural network (DNN)-based approach for the metasurface inverse design [78].

Predicting the desired S-parameters or the complex reflection and transmission coefficients for specific frequency ranges is the most common optimization problem in meta-atom design. The optimization targets can be represented as vectors of discrete sampling points, varying in their density of samples and corresponding spectra, tailored to practical applications. In the conventional design process, extensive full-wave simulations using commercial software such as CST Microwave Studio are typically the most time-intensive step for obtaining the wideband S-parameters or complex reflection/transmission coefficients of a meta-atom. To this end, accelerating the simulation process using deep-learning (DL) methods is a worthwhile endeavor. Specifically, DL approaches often design various artificial neural network (ANN) architectures to learn the relationship between the structural parameters of a meta-atom and its discrete electromagnetic (EM) responses. Due to the intrinsic black-box nature of the learning process in ANNs, their network structure provides a high degree of freedom, allowing for the utilization of various types of convolutional neural networks (CNNs) [79], recurrent neural networks (RNNs) [80], or transformers [81]. Empirically, the prediction accuracy of ANNs is primarily determined by the quality and quantity of the training dataset. Additionally, the network structure of ANNs dictates how close the prediction accuracy can approach its upper limit. Hence, it is advisable to experiment with diverse ANN architectures, varying the number of layers/nodes, activation functions, and inter-layer connections, until

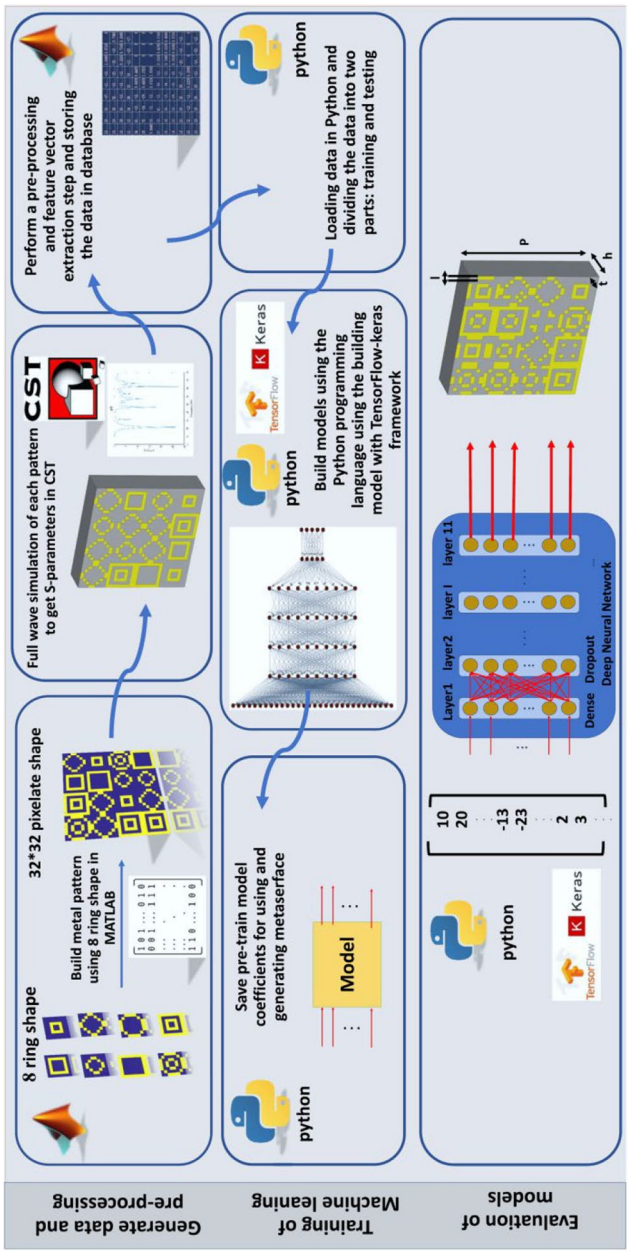


Fig. 1 Sketch representation of the design process of DNN-based approach for metasurface inverse design, which commonly includes three steps: data simulation and pre-processing, network training, and network evaluation [78].

achieving the desired prediction accuracy. Because the suitable ANN structure depends on specific problems and training datasets, our discussion will not emphasize the network structure design. Instead, it will focus on parametrization methods for meta-atom geometry and functionality, along with their integration into the overall intelligent design process using inverse design methods.

Intelligent algorithms for meta-array design and optimization

Programmable metasurfaces allow dynamic manipulation of EM waves in both temporal and spatial dimensions by adjusting the state of unit cells using digital codes. The traditional optimization algorithms, such as genetic algorithms and particle swarm optimization, are computationally demanding, making it challenging to compute optimal codes for metasurfaces in real-time. Shan et al. proposed a fully convolutional neural network (termed ConvNet) to code the programmable metasurface for multi-beam steering, which combines the efficiency of back-projection methods and the accuracy of nonlinear optimization techniques, making it suitable for unique applications that require the dynamic and complex beamforming [82]. The design process of optical metasurfaces typically requires numerous forward simulations to optimize their performance, which can be computationally intensive and time-consuming. Traditional methods like solving Maxwell's equations on a meshed grid or using Mie-scattering solutions have limitations in terms of computational efficiency, memory usage, and geometric constraints. Zhelyeznyakov et al. presented a data-driven forward simulation framework for a faster and more efficient inverse design of optical metasurfaces [83]. As shown in Fig. 2(a), this approach employs Singular Value Decomposition (SVD) to obtain low-dimensional representations of the scattered field data. Then, a differentiable neural network model is created to map the input geometries and configurations of the metasurface scatterers to the low-dimensional representations of the output field obtained from the SVD. As a result, it achieves more accurate results compared to methods based on the local phase approximation, up to 10^4 times faster, and requires 15 times less memory than mesh-based solvers. This work establishes a high-fidelity and low-cost forward simulator by combining data-dimension reduction with a differentiable neural network, fundamentally alleviating the computational bottleneck of full-wave simulations in inverse designs of optical metasurfaces. Its key insight lies in learning and compressing the essential features of scattered fields rather than replacing the physical solver, enabling orders-of-magnitude acceleration and memory reductions while preserving the accuracy, thus paving the way for rapid iteration and large-scale optimization of complex metasurfaces.

Currently, most efforts to develop multifunctional metasurfaces focus on empirically accommodating more functionalities, simply ignoring the intrinsic limitations of the meta-atoms, which inevitably increases structural complexity yet results in the ultimate design constraints. Ma et al. proposed to embed machine-learning models in both gradient-based and non-gradient optimization loops for the automatic implementation of multifunctional metasurfaces [85]. Different from traditional methods, this approach integrates phase retrieval and meta-atom structural design into a single process, which can capture the statistical relationships among meta-atom structures and their optical responses, allowing full exploitation of the design space. Experiments on metasurface focusing lenses and holograms, implemented through a readily fabricated

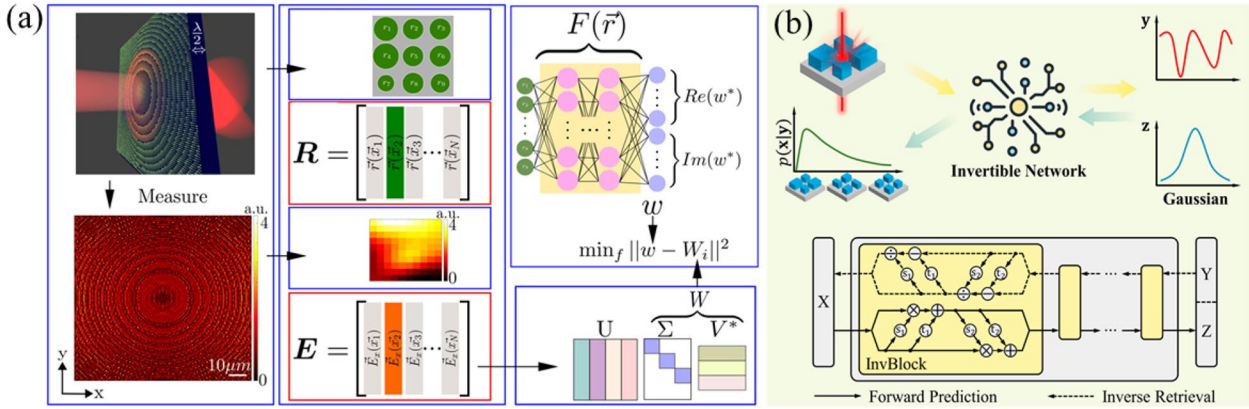


Fig. 2 Deep learning frameworks for metasurface design. **a** End-to-end deep learning framework with unidirectional optimization [82]. This approach uses Singular Value Decomposition (SVD) to extract low-dimensional representations of the scattered field. A differentiable neural network maps metasurface geometry to these field representations, achieving higher accuracy than local phase approximation methods while operating 10^4 times faster and using 15 times less memory than mesh-based solvers. [83] **b** Invertible neural network (INN) framework with bidirectional optimization [83]. The INN concurrently models both forward and inverse design processes, effectively handling the ill-posed nature of inverse problems where multiple structures may produce similar optical responses. This allows fuller exploration of the design space and improved capture of statistical relationships between structure and performance [84].

single-layer structure, showcase eight controllable responses subjected to different combinations of working frequencies and linear polarization states, significantly surpassing the capabilities of conventional physics-guided approaches. While DL-based approaches have shown impressive accuracy and efficiency, they often overlook the highly ill-posed nature of inverse design, where one optical response can correspond to multiple geometrical configurations, posing an underlying challenge. Taking the probabilistic nature of the inverse design problem, Wang et al. proposed an invertible neural network (INN) to simultaneously model both forward and inverse processes for metasurface design shown in Fig. 2(b) [85]. Unlike other DL-based techniques with unidirectional optimization that require separate models for forward prediction of optical properties and backward retrieval of geometry parameters, INN captures a probabilistic model for the inverse process, enabling recovery of the complete posterior over the parameter space given a specific optical response. This allows the generation of novel designs beyond the training data. By integrating the INN with the angular spectrum method, this work offers an efficient and automated end-to-end metasurface design and evaluation system, making it a general method applicable to various optical inverse design problems. The findings from two design examples of high-efficiency meta-lenses and dual-polarization metasurface holograms showcased the efficacy of the proposed INN-based framework to generate diverse and high-performance designs efficiently, outperforming traditional methods. By modeling the posterior distribution of structural parameters, INN transforms the inverse design from a deterministic one-to-one mapping into a probabilistic exploration of the design space, enabling the generation of multiple feasible and novel results beyond the training dataset and enhancing both creativity and practicality.

To further address the challenges faced by the traditional machine learning methods — namely high computational complexity, lengthy design cycles, and limited practical applicability — Jia et al. proposed a novel framework called knowledge-inherited learning (KIL) in designing large-scale and multifunctional metasurfaces [86]. This approach introduces a hierarchical and modular learning strategy that emulates human design cognition: first acquiring knowledge of fundamental building blocks, and subsequently combining and reusing these modules within more complex architectures. To bridge the gap between microscale learning and macroscale metasurface assembly, KIL employs a layered, multi-stage learning scheme. By transferring modular knowledge across different scales and functionalities, it decomposes the metasurface design problem into manageable sub-tasks and enables the reuse of pretrained models through a process of knowledge inheritance. The core components of the KIL framework include: elementary module learning, knowledge inheritance mechanisms, module integration and collaborative learning, and a seamless design-to-assembly pipeline. Each level in the design hierarchy is modeled by a dedicated neural network tailored to a specific task, allowing complex metasurface functionalities to be broken down into simpler sub-models. This enhances both learning efficiency and interpretability. For multifunctional metasurfaces —such as devices exhibiting simultaneous reflection, absorption, and scattering behaviors—KIL introduces a collaborative coding strategy. This enables the simultaneous optimization of multiple physical parameters without retraining the entire network. Once training is complete, low-level models can be reused across different metasurface designs. For example, an absorption module trained on small-scale unit cells can

be scaled up and integrated into large-area metasurfaces. KIL supports arbitrary stacking and tiling of modules with varying sizes and topological configurations. Moreover, KIL incorporates real-world imperfections, such as fabrication tolerances and material losses, into the learning process. This ensures that the metasurfaces designed through the framework are not only effective in simulations but also robust under practical physical implementations.

The traditional inverse design methods such as topology optimization and machine learning-based approaches often struggle with the issues of spectral ambiguity and non-uniqueness, where multiple distinct structures can produce identical or similar spectral responses. To address these challenges, Chen et al. proposed a Generation-Elimination (GE) framework [87], which first generates a diverse set of candidate structures and then eliminates those that fail to meet the target spectral requirements. This framework adopts a dual-phase strategy comprising a Generation Phase and an Elimination Phase. In the Generation Phase, a generative model is employed to produce a wide range of candidate metasurface structures. The model is trained to learn the associations between target spectra and feasible geometric configurations. During the Elimination Phase, a separate evaluation model assesses the generated candidates by predicting their actual spectral responses and retaining only those structures whose predicted spectra closely match the desired targets. The entire process relies on the collaborative functioning of a spectra-to-structure generator and a structure-to-spectra predictor. To support this framework, the authors constructed a large-scale dataset consisting of 200,000 metasurface patterns and their corresponding spectra using full-wave electromagnetic simulations. Each pattern was designed within a square periodic unit cell and represented as a binary structural layout. Both models were trained via supervised learning, with loss functions specifically tailored to enhance spectral fidelity. The generator was optimized to maximize the likelihood of generating valid designs for a given spectral input, while the forward model was trained to minimize the prediction error of spectral responses. Once trained, the GE framework demonstrated high accuracy in predicting spectral responses from structural designs and proved capable of generating a diverse set of structures that closely match the desired spectral characteristics.

Benefitting from the precise manipulation of the phase, amplitude and polarization of electromagnetic waves, holographic metasurfaces achieve superior image reconstruction compared to traditional methods [88–90]. To achieve the desired goals, metasurfaces must be meticulously designed to produce specific optical responses. However, conventional inverse design of meta-unit states for holographic imaging by coding metasurfaces primarily relies on computationally intensive gradient-based techniques like the adjoint method or heuristic approaches such as genetic algorithms, making them impractical for large designs. Very recently, the DL-based paradigms capable of acquiring the relationship between complex nanostructure geometry and electromagnetic responses have revolutionized the design of holographic metasurfaces. Various network architectures, including convolutional neural networks (CNNs) [91, 92], generative adversarial networks (GANs) [93], and variational autoencoders (VAEs) [94], have been applied to this field.

Here, we present three representative DL-based learning paradigms: supervised learning, unsupervised learning, and hybrid supervised-unsupervised learning.

Complex-amplitude holographic metasurfaces (CAHMs), offering flexible control over phase and amplitude, have been utilized to manipulate wavefront propagation with exceptional precision, demonstrating superior image reconstruction quality. Traditional design algorithms for CAHMs, based on theories like Huygens-Fresnel and Rayleigh-Sommerfeld diffraction, are computationally expensive due to the need for meta-atom optimization, numerical simulations, and experimental verifications. Zhu et al. presented a novel approach for the design of CAHMs using a residual encoder-decoder convolutional neural network, which directly maps electric field distributions to the input images needed for metasurface design [94]. As detailed in Fig. 3(a), the model is firstly trained using electric field distributions calculated via Rayleigh-Sommerfeld diffraction theory, followed by further refinement through a transfer learning framework to map simulated electric field distributions and input images. The proposed method offers an efficient and accurate way to design CAHMs by focusing on the overall pattern of the metasurface instead of the detailed characteristics of individual meta-atoms, simplifying the design process. Despite the significant advancements, most DL-based methods primarily depend on supervised learning. This implies that training ANNs requires a substantial number of paired samples, typically obtained through extensive full-wave simulations or experimental measurements, which is time-consuming and labor-intensive, posing a significant challenge for researchers.

To overcome this limitation, some unsupervised DL methods have been proposed, which commonly integrate the physical constraints into the network, offering a real-time and robust solution [94]. Liu et al. developed an unsupervised framework based on a conditional generative adversarial network (cGAN) [93]. This approach integrates the physical mechanisms of electric-field distribution and metasurfaces into the model, allowing for the real-time design of the corresponding hologram for a 1-bit coding metasurface when a target holographic image is provided, as shown in Fig. 3(b). Specifically, this method employs electromagnetic (EM) propagation formulas to construct a forward mapping from the metasurface hologram to the corresponding holographic image, allowing the generator responsible for inverse design to be trained using an unsupervised GAN structure. The generator was trained to ensure that the output image closely matches the input image, allowing for the use of arbitrary images to train the generator without needing to prepare their corresponding holograms. While the discriminator was incorporated into the training process to ensure that the output images from the generator were more topologically similar to the input images. Experimental results demonstrate that the metasurface holograms produced by the trained generator exhibit superior imaging quality compared to those generated by the conventional GS algorithm. This work overcomes the dependence on large paired datasets by integrating physical constraints into an unsupervised cGAN. Embedding a differentiable forward model into the generator allows for the inverse design of high-quality holograms guided solely by the physical laws and adversarial training, advancing the design paradigm from data-driven to physics-and-data co-driven. It is anticipated that additional physics-assisted deep-learning methods will be developed for the intelligent design of entire metasurfaces, which will improve the interpretability and generalization capabilities of ANNs. In Fig. 3(c), Xi et al. presented a hybrid supervised-unsupervised neural network for the design and generation of complex polarization holograms, which directly derives

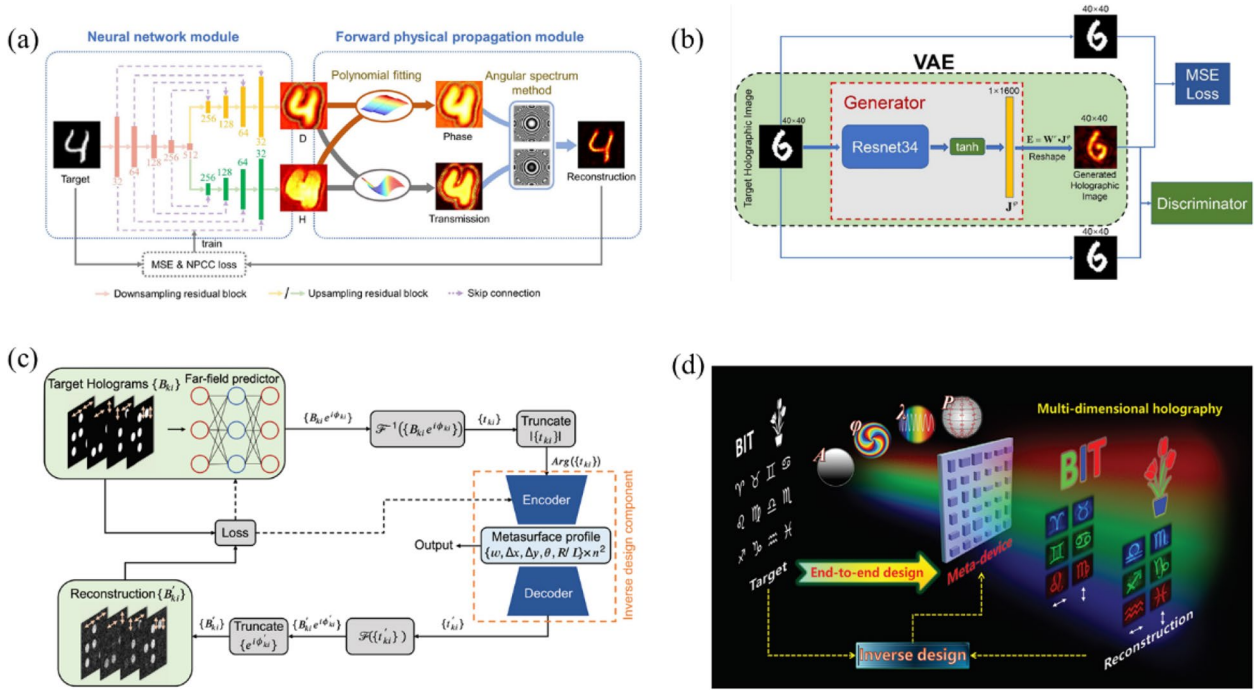


Fig. 3 Intelligent design of holographic metasurface. **a** Schematic diagram of intelligent complex- amplitude holographic metasurfaces design. A residual encoder-decoder convolutional neural network is initially pre-trained using electric field distributions calculated by diffraction theory, followed by further refinement through a transfer learning framework to map simulated electric field distributions and input images [94]. **b** Illustration of physics-assisted unsupervised GAN for real-time holography. The VAE structure includes the generator and the EM propagation process, with the dyadic Green's function (DGF) serving as the physical prior layer. A discriminator was implemented to improve the imaging quality of the generator [93]. **c** Integrated deep neural network to design metasurface polarization hologram. The decoder is pre-trained with supervised learning, while the far-field predictor network and the encoder network are trained with unsupervised learning [95]. **d** End-to-end inverse design framework for the design of multi-dimensional metasurface holography [96].

metasurface structural profiles from holograms, accommodating a variety of polarizations and supporting up to four different co- and cross-polarization conversion channels [95]. Importantly, this method focuses on leveraging information directly from the holograms rather than relying on detailed physical knowledge of the constraints, enabling the design process to be more efficient and accessible.

Another inverse design framework is the end-to-end inverse design framework for the design of multi-dimensional metasurface holography (Fig. 3(d)) [96]. This framework directly connects the metasurface to reconstructed images and utilizes a loss function to guide the metasurface update. As a result, the need for hologram calculation is eliminated, resulting in a greatly simplified design process. The end-to-end inverse design framework proposes a method to convert physical design problems into mathematical optimization problems. A meta-atom library is used as the solution space, with amplitude and phase response, while known parameters include wavelength, reconstruction distance, polarization state, and target image. The loss function is defined as the difference between the reconstructed image and target image. This approach establishes a direct relationship between the metasurface's structural parameters and the output reconstructed images, enabling the end-to-end design of meta-devices. By changing the target performance requirements, the exact metasurface parameters can be quickly obtained using the proposed algorithm. The proposed end-to-end inverse design method directly connects the device parameter end with the output performance end, simplifying the design process of metasurfaces and facilitating rapid design of meta-devices. This method is conducive to promoting intelligent design of multifunctional components and is expected to accelerate the application of meta-devices in areas such as colorful displays, imaging, and storage.

Despite the promising advances in intelligent algorithms for meta-array designs, a persistent and significant challenge lies in the efficient handling of large-scale and high-degree-of-freedom (high-DoF) optimization problems. As the metasurfaces evolve towards larger apertures (involving thousands to millions of unit cells) and more sophisticated multifunctions (simultaneously controlling amplitude, phase, polarization, and frequency across multiple channels), the design space grows exponentially, leading to a combinatorial explosion. The traditional gradient-based optimization often struggles with the non-convex and high-dimensional landscapes, prone to getting trapped in local minima. While the data-driven approaches such as deep learning frameworks offer powerful surrogate models and pattern discovery capabilities, they still face some hurdles: the curse of dimensionality demands prohibitively large and diverse training datasets, and the generalizability of models trained on limited-scale or specific functions remains questionable when scaled to vastly different array sizes or unprecedented performance targets. Further, integrating fabrication constraints (e.g. discrete states and manufacturing tolerances) and real-time reconfiguration requirements for programmable systems into the high-DoF optimization loops adds another layer of complexity. Future research is likely to focus on more hierarchical, modular, and physically constrained learning architectures, as well as hybrid strategies that combine the global search ability of AI with the local precision of traditional optimization, to make the intelligent designs of extremely large-scale and complex meta-systems computationally tractable and practically viable.

Applications of intelligent metasurfaces

The intelligent metasurfaces designed and constructed using the above methods have been widely applied in various areas and scenarios. Here, we highlight three representative proof-of-concept demonstrations in this burgeoning field as follows.

Metasurfaces can be easily concealed within the environment while still providing powerful beam and signal manipulation capabilities for physical-layer wireless attackers. In recent years, the application of metasurfaces has introduced new research opportunities in the field of information security. To address the issue that sophisticated attack strategies at the physical layer in current wireless communication systems leave detectable traces in the physical environment, Wei et al. have developed a metasurface-enabled smart wireless attacks at the physical layer operating in the Wi-Fi band at around 2.4 GHz [97], supporting both active and passive operation modes as shown in Figs. 4(a) and 5(a). In the passive mode, the attacker can eavesdrop on the wireless information transfer of a target by controlling the programmable metasurface without actively radiating any signal. In the active mode, the attacker can both eavesdrop and falsify wireless communications by sending deceptive information to the target. Both two operational modes can realize the minimality of the attacker's detectability. They experimentally showed that programmable metasurfaces can enable sophisticated attacks, ranging from conventional eavesdropping to the disruption of communication and information falsification. The proposed system reveals dual-edged potentials of the intelligent metasurfaces in physical-layer wireless security, demonstrating how they can enable the stealthy eavesdropping, jamming, and spoofing attacks with the minimal active signal emission. It expands the battlefield of information security from the digital domain to physical domain, necessitating a re-evaluation of security architectures for next-generation communication systems.

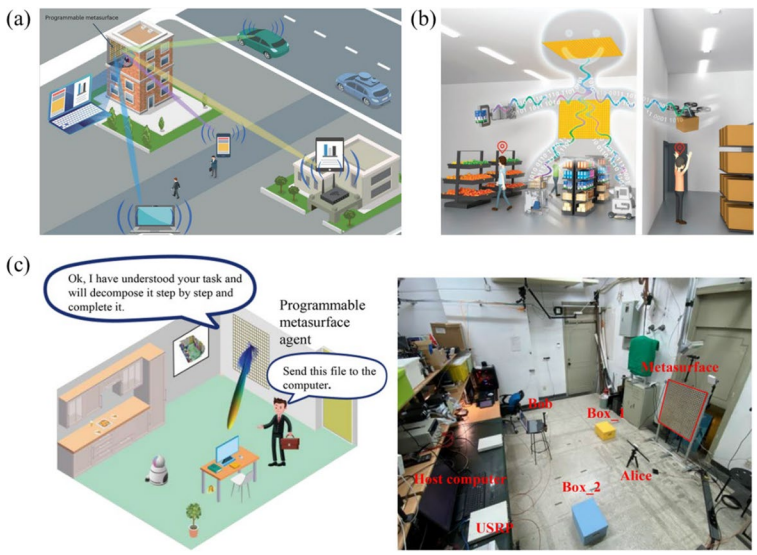


Fig. 4 Conceptual illustrations of three representative application scenarios based on intelligent metasurface systems. **a** Metasurface-enabled smart wireless attacker at the physical layer for hacking a wireless local area network in a typical indoor environment [97]. **b** Intelligent indoor metasurface robotics for smart human–machine interaction [98]. **c** Language-controllable programmable metasurface agent in the context of large language models [99]

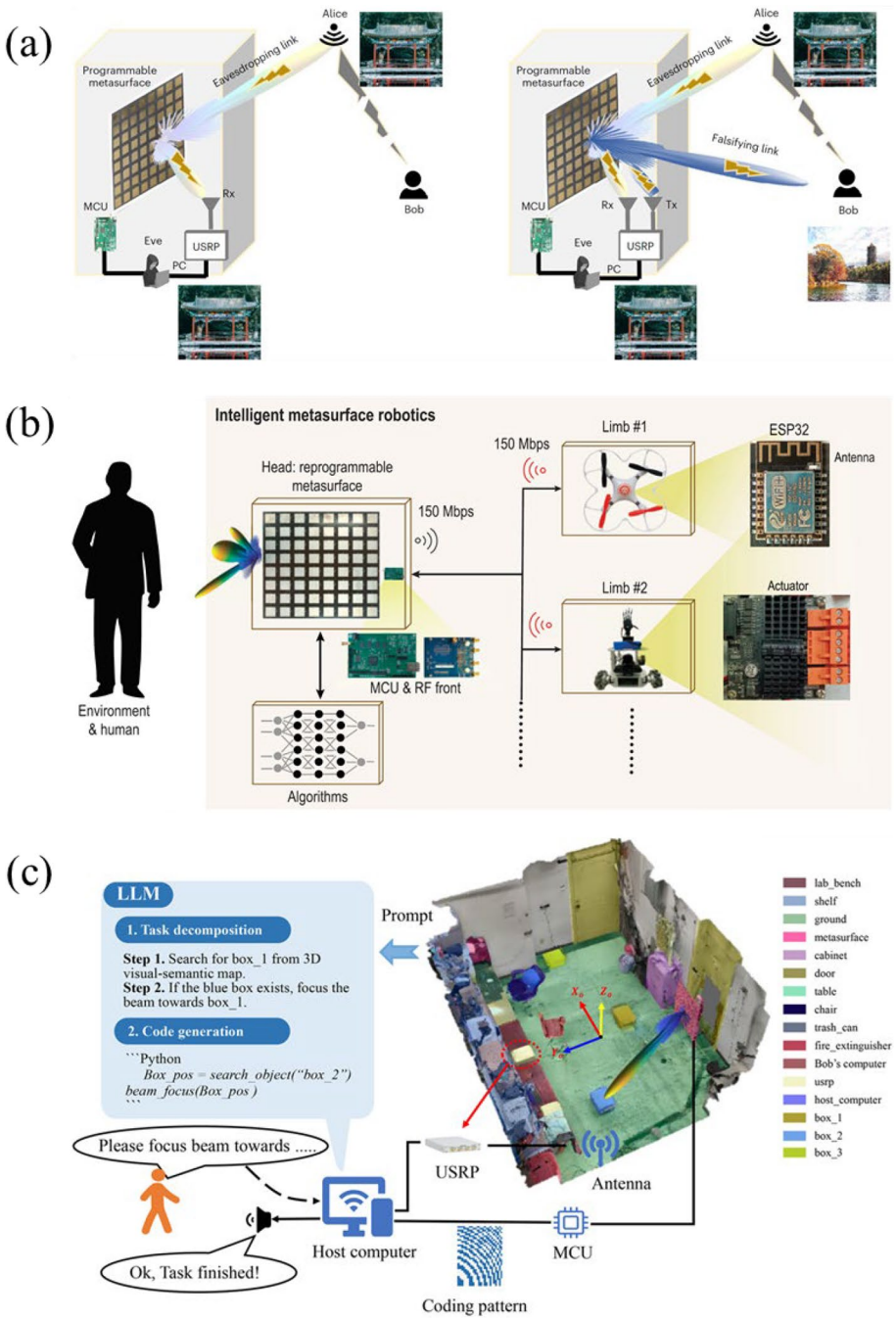


Fig. 5 System configurations of the three representative metasurface-enabled intelligent systems. **a** Metasurface-enabled smart wireless attacker at the physical layer for hacking a wireless local area network in indoor environment under passive mode (left) and active mode (right) [97]. **b** Intelligent indoor metasurface robotics for smart human–machine interaction [98]. **c** Language-controllable programmable metasurface agent for human-level smart EM manipulation together with a 3D visual- semantic map (right) [99]

Based on the successful applications of intelligent metasurfaces in perception, communication, information security and other areas, Li et al. developed a prototype of intelligent human–machine interaction robotic system designed for ambient assisted living and smart healthcare at home [98]. This innovative system, known as intelligent

indoor metasurface robotics (I2MR), integrates metasurface technology, AI-powered data processing, and motorized limbs, as shown in Figs. 4(b) and 5(b). In I2MR, all sensing and computing tasks are centralized in a robotic brain equipped with a metasurface-centered microwave perception modality and an AI-enhanced data processor, while the limbs (such as motorized vehicles and airborne drones) execute instructions wirelessly received from the brain. As the core innovation of the system, the computation-enabled programmable metasurface can flexibly shape microwave propagation in the indoor wireless environment, enabling low-latency and high-resolution three-dimensional imaging and posture detection of humans, even around corners and behind thick concrete walls. This capability forms the basis for the I2MR brain's action decisions, and then it establishes a high- capacity communication link with the robotic limbs to transmit instructions. As a result, I2MR possesses real-time awareness of its dynamic indoor environment, facilitating adaptive human–computer interaction. This work integrates a programmable metasurface which acts as the core sensing and communication hub into a robotic system, enabling non-line-of-sight, through-wall 3D perception and human pose detection. It addresses the limitations of traditional optical sensors in complex indoor environments and offers a powerful non-invasive sensing modality for the applications in smart healthcare and assisted living.

However, it should be noted that the previous systems failed to demonstrate true general intelligence, and often required human intervention to solve specific or simple problems, making it difficult to emulate human-like processes. To address this issue, Hu et al. established the concept of a language-controllable programmable metasurface named as programmable metasurface agent [99]. As illustrated in Figs. 4(c) and 5(c), it leverages the notable capabilities of large language models (LLMs) to achieve human-like intelligence in autonomous electromagnetic manipulations. A visual semantic map is also constructed to facilitate the language-controllable electromagnetic manipulations in the 3D physical environment. They experimentally illustrated the agent's strong ability to autonomously decompose complex electromagnetic tasks into simple and executable steps using the self-planning capability of LLMs, continuously refine its electromagnetic manipulation code through trial and error based on the environmental feedback, explore unknown environments, and collaborate with other agents or humans. This work pioneers the concept of a programmable metasurface agent by leveraging LLMs to imbue the metasurfaces with high-level cognitive and autonomous decision-making capabilities. Using the natural language as an interface, it enables the decomposition of abstract human instructions into executable electromagnetic manipulation steps, marking a step toward embodied intelligent electromagnetic systems.

Intelligent computing based on metasurfaces

The rapid development of artificial intelligence (AI) has significantly improved productivity and facilitated scientific innovation. However, it has also introduced a series of new challenges. It is estimated that training and inference for large-scale models such as GPT-4 require approximately 40 PFLOPS of computational power, leading to enormous energy consumption. Considering that traditional GPU-based computing hardware is constrained by the limitations of semiconductor manufacturing processes, developing new computing hardware with high energy efficiency and high-speed performance

has become a key research focus in intelligent computing. To address this challenge, the concepts of all-optical diffractive neural networks and optical-electronic hybrid neural networks have been proposed, presenting a novel deep learning computing framework. These approaches implemented by various physical platforms have potentials to break through the bottlenecks of traditional computing hardware, offering a new pathway toward efficient, low-power, and large-scale AI systems. This section reviews the progress in this field, which is categorized into three main implementation paradigms: free-space diffractive neural networks, chip-integrated diffractive neural networks, and metasurface-based neural networks.

Free-space diffractive neural networks

Free-space diffractive neural networks utilize the propagation and interference of light in three-dimensional space to perform analog computations. This paradigm offers inherent parallelism and ultra-fast processing speeds, serving as the foundational architecture for optical neural computing.

In 2018, Lin and coworkers proposed the framework of all-optical diffraction deep neural network (D^2NN) [100], using the diffraction of light between layers to replace large-scale matrix computations in the forward propagation process of fully connected networks. A D^2NN is composed of several passive diffractive layers consisting of multiple diffractive units, which can perform a complex-value modulation on the input signal. According to the Huygens-Fresnel diffraction theorem, after receiving the signal from the previous layer, each diffractive unit radiates a secondary wave, whose properties are decided both by the input and the transmission characteristic of its own. Figure 6(a) shows the process of light propagation in D^2NN, indicating that D^2NN can simulate the forward propagation process in the fully connected network, with the diffractive units being regarded as neurons with adjustable complex values. For general classification tasks, D^2NN determines the classification results through detecting the distribution of photon energy on the output plane and Fig. 6(b) shows an example of D^2NN processing data from the MNIST dataset. Ten detectors are used to monitor the photon energy falling in the detection area, and then the prediction result is made based on the largest photon energy. In addition, D^2NN can also be trained as an imaging network, as shown in Fig. 6(c). This seminal work establishes a paradigm of all-optical diffractive deep learning by physically implementing the neural network forward propagation through passive diffraction, mapping the mathematical operations directly onto light-matter interactions for ultra-fast and energy-efficient computations at the speed of light.

Unlike conventional electronic chips, optical computing generates minimal heat and inherently supports parallel processing. Researchers have long been exploring optical methods to construct various artificial neural network structures, enabling ultra-fast information processing for complex tasks. However, due to the high flexibility of artificial neural network designs and the accumulation of system errors, optical processing architectures are often difficult to reconfigure, which limits model complexity and thus constrains experimental performance. To overcome this issue, Zhou et al. proposed a reconfigurable diffractive processing unit (Fig. 6(d)) [50], which is an optical-electronic hybrid computing architecture based on light diffraction. This system supports different neural network configurations and enables high-complexity models with up to millions

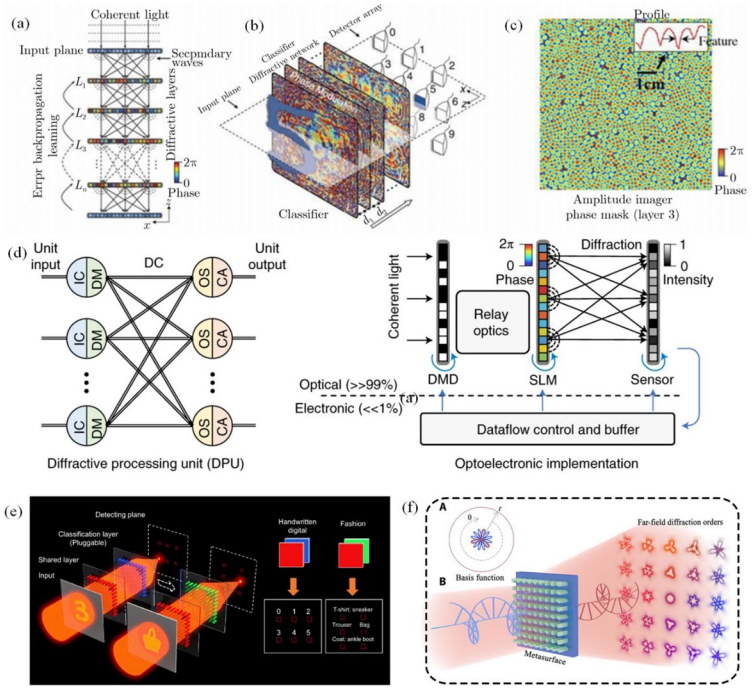


Fig. 6 Free-space diffractive neural networks. **a**-**c** Diffractive deep neural networks (D^2NN). **d** The DPU implemented by a DMD, a phase SLM and a CMOS sensor is a large-scale perceptron-like optoelectronic computing building block that can be programmed to construct different D^2NNs [50]. **e** Concept of pluggable diffractive neural networks (P-DNN) for multiple tasks [101]. **f** Scheme of DVM based on dielectric geometric metasurface [102].

of neurons. In addition to this computing framework, the team also developed an adaptive training method to mitigate system errors. Experimental results demonstrated outstanding accuracy on image and video recognition tasks using benchmark datasets, surpassing the performance of state-of-the-art electronic computing platforms. This system addressed the critical reconfigurability bottleneck in the optical neural networks by introducing a hybrid optical-electronic architecture, successfully balancing the inherent parallelism and speed of light with the flexibility and adaptive control of electronics for general-purpose programmable computing.

Considering that current ONN architecture cannot be easily reconstructed to realize multi-functional artificial intelligence systems simultaneously, He et al. proposed a pluggable diffractive neural network (P-DNN) to meet the recognition demands of multiple datasets (Fig. 6(e)) [101]. By introducing internal plug-ins, this system allows the replacement of task-specific recognition layers while keeping the shared backbone (common layer) unchanged, thereby enabling multi-task recognition and accommodating diverse dataset requirements. This architecture significantly enhances the adaptability and reconfigurability of diffractive neural networks. The proposed P-DNN consists of two types of layers: a common layer for preprocessing input information and a replaceable task-specific classification layer. The work validated the working mechanism of P-DNN using classification tasks on two distinct datasets: MNIST and Fashion-MNIST. The P-DNN framework, combined with a cascaded metasurface design, demonstrates high efficiency and accuracy in classifying diverse datasets, highlighting its potential

value in machine learning applications. These plug-ins exhibit excellent reconfigurability and allow seamless task switching through simple plug-and-play operations. With advantages such as ultra-low energy consumption and ultrafast processing speed, the P-DNN architecture provides enhanced flexibility to optical neural networks. It also shows strong potential for a wide range of practical applications, including intelligent optical filtering in microscopic imaging and real-time object detection in autonomous driving systems.

Drawing on the concept of basic functions in mathematics, Zhang et al. proposed the Dammann vortex metasurfaces (DVM) [102], which is based on a linear combination of some customized basic patterns, to generate diffraction multiplexing and produce generalized vortex beams (GVBs). As is shown in Fig. 6(f), this approach enables the spatial features of each diffraction order to no longer be the traditional donut-shaped vortex beam, but rather exhibit different intensity distributions carrying OAM. DVMs use phase-only modulation to produce the desired intensity profiles of GVBs for various diffraction orders. A superperiod comprises several pixel arrays in the shape of a forked chessboard with optimized phase distributions. This enables DVM to achieve multiplexing/demultiplexing by generating/detecting the spatial distribution of OAMs within the desired diffraction orders. The GVB beams produced exhibit distinct shapes in varying spatial diffraction orders, deviating from the typical toroidal form. This optical representation serves as a valuable tool for mathematical computation. The utilization of such DVMs may present opportunities for the exploration of diverse applications, including vortex information encryption, optical manipulation and communication, and quantum entanglement, among others.

While the free-space architectures demonstrate powerful parallel processing capabilities, their bulky nature and sensitivity to misalignment pose challenges for system integration and scalability. This has driven the field towards more compact and robust chip-integrated implementations, where the optical computing units are miniaturized and embedded into planar substrates to enhance the stability and facilitate the mass production.

Chip-integrated diffractive neural networks

Chip-integrated diffractive neural networks aim to miniaturize the optical computing elements onto planar photonic chips, improving the system stability, scalability, and potential for integration with the electronic components.

In 2023, Chen et al. introduced ACCEL [29], a fully analog computing chip that integrates both electronic and optical computing (Fig. 7(a)). Unlike conventional vision tasks, where sensors first capture high-resolution images, digitize them via numerous analog-to-digital converters (ADCs), and then process them in digital computing units, ACCEL does not rely on digital units for optical computing. Instead, it integrates optical analog computing (OAC) and electronic analog computing (EAC) to achieve scalability, nonlinearity, and flexibility on a single chip. The ACCEL system boasts an energy efficiency of 74.8 tera-operations per watt per second and a computing speed of 4.6 tera-operations per second, with over 99% of computations performed optically. This represents a three-order-of-magnitude improvement in energy efficiency and a one-order-of-magnitude improvement in computing speed over the most advanced

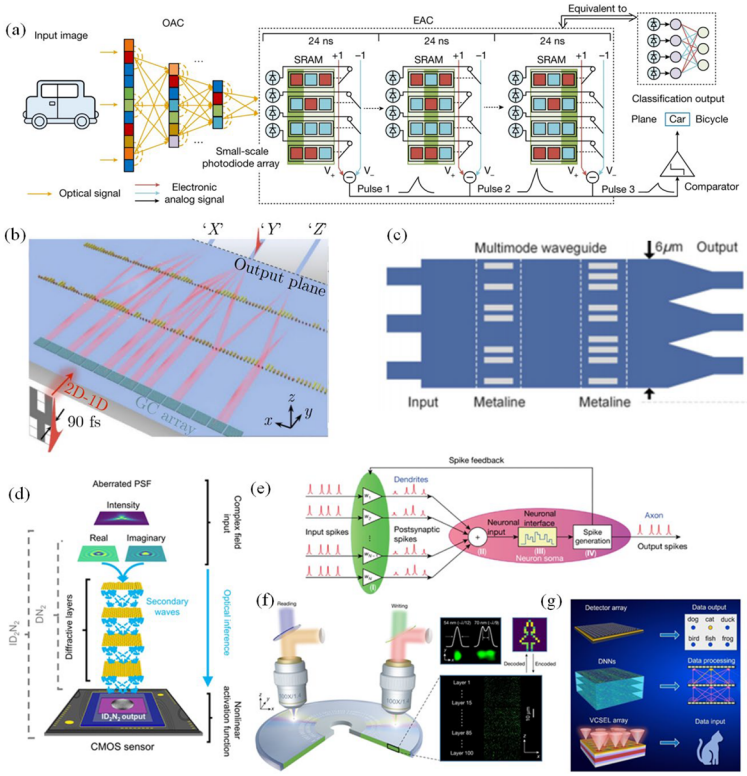


Fig. 7 Chip-integrated diffractive neural networks. **a** The workflow of ACCEL [29]. A diffractive optical computing module processes the input image in the optical domain for feature extraction, and its output light field is used to generate photocurrents by the photodiode array for analog electronic computing directly. EAC outputs sequential pulses corresponding to multiple output nodes of the equivalent network. The binary weights in EAC are reconfigured during each pulse by SRAM, by switching the connection of the photodiodes to either V_+ or V_- lines. The comparator outputs the pulse with the maximum voltage as the predicted result of ACCEL. **b** Schematics of the planar monolithic diffractive neural network [54]. **c** Architecture of multimode DONN [103]. (d) Schematic of diffractive neural network integrated on a commercial CMOS chip performing pupil phase retrieval [104]. **e** Schematic of a photonic network consisting of N pre- synaptic input neurons and one post-synaptic output neuron connected via PCM synapses [105]. **f** Principle of nanoscale optical writing and reading on a blank AIE-DDPR disk [106]. **g** The on-chip integration design of DNNs based on VCSEL arrays [107]

processors available today. Such a breakthrough makes ACCEL highly promising for applications in wearable devices, autonomous driving, and industrial inspection. This breakthrough demonstrates a fully analog optoelectronic chip that seamlessly fuses the optical and electronic analog computing, achieving orders-of-magnitude improvements in the energy efficiency and speed by processing visual information directly in the analog domain, bypassing the power-hungry digital conversions.

The current photonic computing systems are limited to relatively simple pattern classification tasks due to their small network size. A common method to increase the number of neurons in a neural network is adding more layers, known as deep learning. However, such a method cannot be applied in photonic computing systems, as the propagation noise is accumulated by layer, ultimately showing an exponential increase, while the computational performance does not improve linearly with the addition of layers. To address this issue and expand the scale of photonic computing systems, a large-scale photonic chip named as “Taichi” is proposed based on breadth

computation, which has been proved to have the capability to solve realistic problems [30]. Compared to the traditional deep learning methods, Taichi employs a multi-cluster approach to expand the network scale. A complex task is divided into several simpler subtasks, each of which is parallel processed by a series of independent clusters, consisting of multiple basic on-chip execution units for Taichi (TEUs).

Since diffractive neural network architectures based on phase surfaces typically require bulky components and struggle to mimic the multitasking capabilities of the human brain, Luo et al. proposed a multi-skill diffractive neural network (MDNN) based on metasurface devices [108]. This system enables chip-scale multi-channel sensing and multitasking within the visible light spectrum. By leveraging polarization multiplexing in subwavelength nanostructures, the team developed a multi-channel classifier capable of simultaneously recognizing digit characters and fashion items. The artificial neuron density reaches 6.25×10^6 mm⁻2, with further scalability depending on the number of channels. Additionally, this metasurface device can be integrated with complementary metal–oxide–semiconductor (CMOS) imaging sensors, enabling a chip-level architecture that performs physical-layer information processing. This design significantly enhances the energy efficiency and speed of image processing, making it suitable for machine vision, autonomous driving, and precision medicine applications. In 2022, Gu et al. reported a planar monolithic diffractive neural network [54], employing a system composed of numerous nanoscale phase shifters for image classification tasks. As illustrated in Fig. 7(b), two subwavelength rectangular grooves function as a phase shifter, with groove length serving as a trainable parameter that adjusts the network weights during training. The system employs gradient descent for optimization, demonstrating its feasibility in training deep learning models. These works represent a crucial shift towards ultra-compact and chip-integrated diffractive neural networks, using the metasurfaces or nanoscale phase shifters to achieve high neuron density and on-chip parallel processing, which is foundational for developing portable and low-power intelligent vision systems.

In 2023, Chen et al. introduced a monolithic diffractive optical neural network (DONN) based on a one-dimensional metasurface [53]. In this design, the hidden layers consist of a series of neuron units, where each unit is a set of silicon dioxide-filled grooves. By training a planar single- hidden-layer DONN using forward and error backpropagation algorithms, the system successfully performed classification tasks. Furthermore, increasing the number of hidden layers and implementing algorithmic compensation for system errors further enhanced classification performance. For DONNs based on diffractive analysis methods, accurately modeling the evolution of optical fields within the structure remains challenging, and real-world applications are often limited by open-boundary losses. To address this issue, as shown in Fig. 7(c), the team proposed a multi-mode DONN architecture based on an eigenmode analysis method [103]. This approach enabled the construction of a generalized library of input, output, and metallic waveguide structures, leading to the implementation of a multi-mode DONN based on this structural library. This series of work advances the chip-scale diffractive networks by exploring various architectural strategies (from 1D meta-grooves to multi-mode eigenmode analysis) to model and control on-chip light

propagation more accurately, addressing key challenges in scalability and loss for the integrated photonic computing.

Retrieving the phase distribution of an incoming wavefront is vital for imaging systems across all scales. The phase distribution across a wavefront can be conveniently described by Zernike polynomials, which provide a complete, orthogonal basis over a unit circle, facilitating the mathematical description of arbitrary pupil phase distributions with minimal variance. Various phase retrieval methods, including Gerchberg-Saxton and techniques like modal wavefront sensing and pupil segmentation, have been developed. Recently, deep learning and Artificial Neural Networks (ANNs) have been used for both direct and indirect phase retrieval. In 2022, Gu et al. proposed a new single-step method for retrieving the pupil phase from an optical beam's Point Spread Function (PSF) using a diffractive neural network (DN_2) and an imaging sensor [104]. The DN_2, operating in the near-infrared (NIR) wavelength, passively reconstructs an intensity distribution representing the pupil phase. This distribution is detected by the sensor, providing the nonlinear response needed for an integrated diffractive deep neural network (ID_2N_2) module (Fig. 7(d)). An experimental setup was designed to demonstrate the capability of the ID_2N_2 to retrieve PSF phases and map them into intensity distributions. The compatibility of diffractive elements with CMOS sensors indicates the capability for directly obtaining pupil phase information from a detector module, eliminating the need for further digital post-processing.

The limitations of the von Neumann architecture, particularly in terms of data transfer speed and energy efficiency bottlenecks (known as the memory wall), have restricted its performance improvements. In contrast, neuromorphic models offer bio-inspired advantages by emulating the behavior of biological neural networks, thus achieving more efficient information processing and learning mechanisms (Fig. 7(e)) [105]. Advancements in early works and large-scale neuromorphic systems have demonstrated the potential of neuromorphic hardware in handling complex tasks [50, 100]. Its promise of speed, intelligent performance, and energy efficiency positions neuromorphic computing as a leading candidate for next-generation computing. Memristors can simulate the plasticity of biological synapses and are key components for achieving low-power, high-efficiency neuromorphic computing. The advancement of photonic memristors using special materials like VO_2 offers promising opportunities for enhancing neuromorphic computing through various fabrication techniques. However, some challenges persist in the practical realization of these devices. Integrating single components into complex, large-scale, and possibly three-dimensional (3D) neuromorphic structures remains difficult. Techniques for joining or assembling these components may introduce stress and defects. Additionally, achieving nonlinearity in the systems fabricated using the techniques like laser nano-lithography can be challenging due to the limited availability of high refractive index photoresists. Overcoming these fabrication and integration challenges is crucial to realize their full capabilities. Nevertheless, this field is also filled with opportunities. The successful realization of all-optical neuromorphic hardware will bring revolutionary improvements in the computational efficiency and energy reduction.

Optical Data Storage (ODS) provides a cost-effective and durable solution for long-term data archiving compared to current data storage technologies such as semiconductor flash devices and hard disk drives (HDDs). Zhao et al. introduced an innovative

approach to significantly enhance ODS capacity by transitioning from planar to 3D storage, utilizing multiple layers and surpassing the optical diffraction limit (Fig. 7(f)) [106]. This is achieved through the development of a dye-doped photoresist film with aggregation- induced emission (AIE) properties, which allows for optical stimulation via femtosecond laser beams. The AIE phenomenon provides a unique storage mechanism, facilitating super-resolution scaling and enabling the creation of petabit and even exabit-level storage capacities through stacking nanoscale disks. Given the inherent advantages of DNN in processing 2D data directly and parallelly, Gu et al. investigated the potential of using addressable vertical-cavity surface-emitting laser (VCSEL) arrays as an innovative data input mechanism [107]. They propose employing these arrays as an integrated foundation for constructing compact and active DNNs tailored for next-generation on-chip vertically stacked photonic neural networks, as shown in Fig. 7(g). By leveraging the capabilities of VCSEL arrays, the realization of micron-scale three-dimensional photonic chips with modulation bandwidths reaching tens of gigahertz is anticipated.

The chip integration can enhance the stability, but it remains difficult to achieve dynamic programmability, efficient nonlinear activation, and deep network topology within the strict planar confines. Metasurfaces with their unprecedented ability to arbitrarily tailor electromagnetic wavefronts in an ultrathin profile emerge as a revolutionary platform to construct the reconfigurable, multifunctional, and highly integrated neural networks directly at the hardware level, opening a distinct and complementary pathway.

Metasurface-based neural networks

Metasurface-based neural networks leverage the subwavelength-scale and programmable elements of metasurfaces to act as artificial neurons or interconnects, enabling novel neuromorphic functionalities within extremely compact form factors and across diverse electromagnetic spectra.

Beyond the dominant approach of integrating active components, an alternative pathway known as structural programmability is emerging. This approach reconfigures the optical responses through passive but physically tunable metastructures, such as by precise rotations or twisting of stacked metasurface layers, enabling functions like high-capacity optical storage without requiring dynamic electrical signals during the operation [109]. Programmability remains a critical bottleneck hindering the widespread application of artificial intelligence (AI) computing hardware. In the microwave regime, mature device and module designs have made it feasible to implement programmable diffractive neural networks based on metasurfaces. In 2023, Cui proposed a programmable diffractive neural network based on multilayer information metasurfaces, referred to as the Programmable Artificial Intelligence Machine (PAIM) [19]. This architecture demonstrated the ability to perform a range of tasks such as image classification, communication decoding, and reinforcement learning, and it shows promising potential in practical applications, including electromagnetic sensing, real-time multi-beam focusing, wireless communications, signal enhancement, medical imaging, remote control, and the Internet of Things. As illustrated in Fig. 8(a), PAIM is constructed using a set of information metasurfaces, with the multilayer structure serving as the programmable physical layer of a D^2NN. This configuration enables PAIM to be designed as a real-time, retrainable system, wherein artificial neurons are implemented via digital

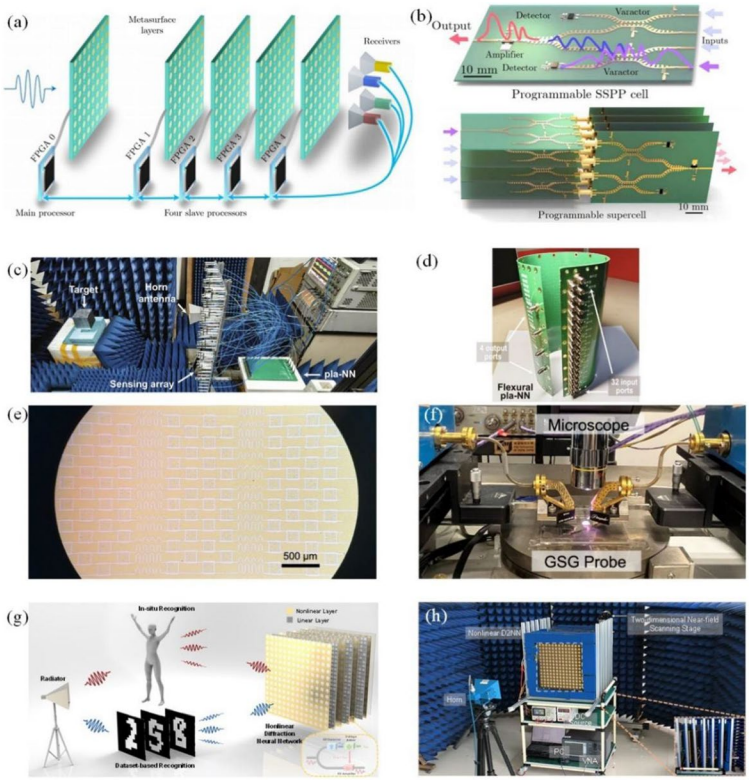


Fig. 8 Metasurface-based neural networks. **a** The assembled FPGA modulation network for PAIM [19]. The receiving antenna array will receive the field distribution of the EM waves, which is processed by FPGAs. FPGAs also guide the update of the bias voltages of the artificial neurons. **b** A single programmable SSPP cell is constructed by integrating the varactors, detectors and amplifier through the SSPP interconnection with four input ports and one output port [18]. A programmable supercell consisting of eight programmable cells (four inputs and four outputs) that can realize 4 × 4 matrix multiplication. **c** Photograph of the object recognition system based on pla-NN. **d** A close-up photo of the fabricated pla-NN curled in a flexible manner [110]. **e** Partial sample display of THz SPNN under the microscope. **f** GSG probe to excite the sample [111]. **g** Schematic of the MN-DNN operation. The network is able to perform both dataset-based image recognition and real-time human posture classification [112]. **h** Experimental test setup of MN-DNN

settings. At the physical layer, PAIM manipulates the energy distribution of transmitted electromagnetic waves through a five-layer information metasurface array, where the amplitude of transmitted waves at each layer can be amplified or attenuated via field-programmable gate arrays (FPGAs). Each metasurface unit cell integrates two amplifiers, and transmission energy is modulated by applying different bias voltages. Thanks to its programmable nature, the first metasurface layer is used as a programmable input mask to encode image information, enhancing both the efficiency of data input and the system's ability to switch between functional tasks. The learning layers consist of four transmissive metasurface layers, forming a fully connected network with programmable weights. These transmission weights can be precisely tuned via FPGA control of the amplifiers' variable gain, effectively realizing a programmable hidden layer. The final output is detected by a receiver, such as a horn antenna or probe, at the output plane. To demonstrate PAIM's intelligent computing capability and versatile programmability, the system was first employed to classify MNIST handwritten digits and oil painting images, as shown in. It was also applied to perform communication encoding and

decoding in the electromagnetic domain. Due to its strong capability in manipulating spatially distributed electromagnetic waves, PAIM is considered a promising solution for implementing spatial division multiplexing (SDM) and enhancing channel capacity. With increasing scarcity in spectral resources, SDM has gained significant attention, and PAIM is anticipated to become one of the key enabling technologies for the future of wireless communication.

As a class of metamaterials, spoof surface plasmon polariton (SSPP) devices have attracted considerable attention due to their flexible dispersion characteristics, strong field confinement, and ease of integration. When varactor diodes are integrated between SSPP waveguides, it becomes possible to dynamically and arbitrarily manipulate SSPP signal waves in real time by programming the bias voltage of the active chips. Consequently, SSPP waveguides show great potential as interconnects in large-scale programmable artificial neural networks. Cui reported a surface plasmon polariton neural network (SPNN) [18] featuring programmable weight factors and nonlinear activation functions. As a fundamental structural unit, a programmable SSPP cell consists of 2 SSPP couplers and 1 power divider (Fig. 8(b)). By controlling the voltage applied to the varactor diodes embedded in the SSPP couplers, the coupling and direct transmission coefficients can be modulated in real time, thereby implementing a programmable weight matrix. Additionally, integrated detectors are used to measure the input signal strength. The detected signal is converted into an output voltage and fed back to a power amplifier. This feedback loop functions similarly to a comparator circuit, with an adjustable reference voltage serving as the threshold for switching between 'on' and 'off' states. This enables the realization of various activation functions, such as ReLU, Tanh, and Sigmoid.

By interconnecting 8 programmable SSPP units, a programmable SSPP supercell with 4 inputs and 4 outputs can be constructed, which forms a fully connected network. The supercell is trained using the classical backpropagation algorithm combined with the ReLU activation function, and the corresponding weight matrix between inputs and outputs is experimentally obtained. Moreover, the SSPP supercell can be used as a replacement for traditional baseband processing units (such as analog-to-digital converters and mixers) to perform image decoding and reconstruction directly on the SSPP platform, enabling image recognition tasks. More importantly, since the decoding process in an SPNN-based communication system takes place entirely in the electromagnetic domain, it is not limited by the clock rate of conventional baseband processors and, in theory, can approach the speed of light. By arranging and stacking multiple SSPP supercells, both fully connected and partially connected neural networks can be constructed. In this work, a partially connected SPNN composed of multiple programmable SSPP supercells was designed and successfully applied to handwritten digit classification. Building on this foundation, the programmable SSPP supercell shows promise for constructing deeper and more complex neural networks, such as convolutional neural networks (CNNs) or recurrent neural networks (RNNs).

To address the challenges of misalignment and large physical dimensions in existing diffractive neural networks, Gu et al. proposed a highly integrated and conformal planar diffractive neural network (pla-NN) [110]. The pla-NN consists of propagation layers and modulation layers, which are constructed from couplers and phase shifters,

respectively. Through training algorithms, the phase distribution within the modulation layers is optimized to establish a mapping relationship between input and output electromagnetic signals. This network performs signal processing directly during electromagnetic wave propagation, inheriting the ultrafast computation speed and energy efficiency of all-optical diffractive neural networks. Fabricated using printed circuit board (PCB) manufacturing techniques, the pla-NN effectively eliminates misalignment issues while offering flexibility for scalable and stackable designs. Numerical simulations were conducted on the Fashion-MNIST dataset using three different data injection methods—direct injection, Fourier-space injection, and hybrid propagation injection—all achieving high classification accuracy, thereby validating the classification capability of the pla-NN architecture. In addition, full-wave simulations were performed using commercial electromagnetic simulation software to further verify the network's performance. Based on the aforementioned model, the overall system is established, as shown in Fig. 8(c). Besides, pla-NN curled in a flexible manner is fabricated (Fig. 8(d)). By integrating the pla-NN with advanced dynamic control techniques and flexible topologies, this architecture holds great potential in high-performance computing, wireless sensing, and flexible wearable electronics. It offers a simple, low-power, and direct signal processing solution for a wide range of applications in the electromagnetic spectrum, including wireless communication, remote sensing, and radar imaging.

Building upon the above research, Gao et al. proposed a terahertz SPNN based on a planar diffractive platform [111], as shown in Fig. 8(e). This network consists of multiple hidden layers, integrating SSPP diffractive layers and phase modulation layers to enable terahertz wave processing on a planar substrate. Each diffractive layer is composed of cascaded compact SSPP coupler arrays, allowing the number of couplers to be flexibly expanded in both horizontal and vertical directions, thereby equipping the structure with excellent scalability. By training the network, the required phase distributions in the phase modulation layers can be obtained, leading to the construction of a complete terahertz SPNN architecture. This work demonstrated the network's capability in classifying basis vectors, recognizing multi-user information, and performing MNIST handwritten digit classification. The measurement setup is shown in Fig. 8(f). Moreover, it validated the network's ability to conduct in-situ information processing of different radiation targets within a wireless transceiver framework. Notably, an input structure for image data was designed and fabricated, comprising a metal grating array, transmitters, receivers, and the SPNN, enabling direct execution of handwritten digit classification tasks. This study expands the application of terahertz plasmonic metamaterials and lays a solid foundation for terahertz on-chip integration, intelligent communication, and advanced computing systems.

To address the challenges posed by nonlinear activation functions in diffractive neural networks, such as latency issues, high power consumption, and cascading complexity, Ning et al. developed a programmable multilayer full-space nonlinear neural network operating in the microwave band, as shown in Fig. 8(g) [112]. The nonlinear layers are constructed using programmable metasurfaces integrated with RF amplifiers, detectors, and voltage adders to implement a ReLU-like nonlinear activation function. This design achieves nanosecond-scale latency, representing an order-of-magnitude improvement in speed compared to photoelectric conversion-based nonlinearities. Moreover, the

system exhibits exceptionally low thresholds and supports reconfigurable nonlinear activation functions. In image classification tasks, including MNIST and Fashion-MNIST, the network achieves competitive accuracy, surpassing conventional linear networks by more than 4%, as illustrated in Fig. 8(h). Furthermore, the proposed architecture enables direct and real-time classification of various human postures in both static and dynamic scenarios. With its low latency and flexible nonlinear activation capability, this architecture tackles a core challenge in analog networks by implementing a programmable and nanosecond-speed nonlinear activation function using the metasurfaces and RF components, enabling high-accuracy and real-time classification of dynamic scenes and proving the essential for closing the performance gap with the digital deep learning.

7.28 TOPS/WQuantitative cross-architecture performance comparison

A comprehensive and quantitative evaluation of computing performance is indispensable for assessing the viability and potential of the emerging intelligent computing paradigms. This section provides a cross-architecture comparison based on key metrics: computational throughput, energy efficiency, latency, and physical footprint, across three dominant categories: the conventional electronic processors, integrated photonic systems, and metasurface-based architectures. The summarized data in Table 1 reveal the distinct performance profiles and design philosophies inherent to each approach.

The electronic processors such as high-performance CPUs and GPUs establish the baseline for programmability and raw throughput in the digital computing. As shown in Table 1(a), the platforms like NVIDIA H100 GPU can deliver up to 19.5 TOPS, enabling complex and general-purpose computations. However, this capability comes at a significant energy cost, with the energy efficiency typically on the order of 10^{-2} GOPS/W. The performance is fundamentally clock-limited and incurs substantial power dissipation and heat generation, highlighting the von Neumann bottleneck that motivates the search for alternative computing paradigms.

The integrated photonic and hybrid optoelectronic systems demonstrate a transformative leap in the energy-efficient and task-specific computation. Architectures like ACCEL and diffractive processing unit (D-RNN), as detailed in Table 1(b), leverage the speed of light and analog signal processing to achieve the exceptional system-level efficiency. For instance, ACCEL reports an energy efficiency of 7.48×10^4 TOPS/W for ImageNet classification, which is several orders of magnitude higher than the electronic counterparts. This phenomenal efficiency stems from performing computations directly in the optical domain with the minimal analog-to-digital conversions. However, these systems often trade off the general programmability for optimized performance on specific tasks (e.g. image classification), and their reported throughput (e.g. 5.95×10^2 TOPS for ACCEL) is deeply tied to these configured tasks.

The metasurface and diffractive neural computing systems prioritize the physical-layer integration, ultra-fast inference at the speed of light, and novel form factors. The data in Table 1(c) illustrate this paradigm. The free-space diffractive neural networks (D^2NN) and planar PCB-based systems exploit light-speed propagation to achieve inference latency on the order of optical transit time, ideal for real-time signal processing. Their computational throughput (e.g. 64.4 TOPS for D^2NN) is realized through massive spatial parallelism. The programmable systems like PAIM introduce

Table 1 Quantitative comparison of computing performance across electronic, photonic, and metasurface-based architectures

(a) Electronic and Digital Computing Platforms

Architecture	Platform	Computing Capacity	Energy Efficiency	Footprint	Latency	Remarks	Ref
CPU	AMD EPYC 9965	576 GOPS	1.15×10^{-3} GOPS/W	~1800 mm^2	Clock-limited	High programmability, low efficiency	-
GPU	NVIDIA H100	19.5 TOPS	0.0485 GOPS/W	814 mm^2	Clock-limited	High throughput, high power	-

(b) Integrated Photonic and Hybrid Optoelectronic Systems

Architecture	Platform	Computing Capacity	Energy Efficiency	Latency	Integration	Task/ Notes	Ref
ACCEL	Hybrid opto-electronic chip	5.95×10^2 TOPS (10-class MNIST) 4.55×10^3 TOPS (3-class ImageNet)	9.49×10^3 TOPS/ W7.48×10^4 TOPS/W	-	Fully integrated	System-level metrics, task-specific output	[29]
D-RNN	Diffractive Processing Unit (DPU)	240.1 TOPS	2548 TOPS/J (direct)1.578 TOPS/J (system)	Optical-scale	Hybrid	Higher node density improves throughput	[50]
Chip-integrated DONN	TDONN	217.6 TOPS	7.28 TOPS/W	~30 ps	PIC	Limited nonlinear depth	[103]

(c) Metasurface / Diffractive Neural Computing Systems

Architecture	Platform	Computing Capacity	Energy Efficiency	Propagation/ Response Time	Footprint	Working Frequency	Ref
Pla- NN	PCB-based	4.4 TOPS	-	Optical	6.6×10^4 mm^2	10 GHz	[110]
Free-space D^2NN	Cascaded diffractive layers	64.4 TOPS	0.25 TOPS/W	Light-speed	1.45×10^8 mm^3	5.8 GHz	[112]
PAIM	Programmable analog information metasurface	30.7 TOPS	0.48 TOPS/W	~1.67 ns (propagation)	1.6×10^8 mm^3 (3D)	5.5 GHz	[19]
SPNN (SSPP-based)	Metasurface + amplifiers	-	-	615 ps (EM) ~ 10 μs (activation)	Planar	Microwave	[18]

OPS/TOPS denote operation-level throughput. Energy efficiency is reported as OPS/W or TOPS/W as provided in the original references. "System-level" indicates inclusion of peripheral electronics

reconfigurability to this domain, enabling diverse functions from classification to wireless encoding, albeit with a current efficiency (0.48 TOPS/W) that bridges the photonic and electronic realms. These architectures excel in the scenarios demanding direct processing of electromagnetic waves, sensor-level integration, or unconventional form factors, though their absolute TOPS values can be lower than that in the integrated photonic chips due to the differences in system scaling and energy delivery.

As summarized in Table 1, the electronic processors dominate in the programmability but suffer from the limited energy efficiency. The integrated photonic accelerators such as ACCEL and D-RNN achieve exceptionally high system-level throughput under task-specific configurations, while the metasurface-based systems emphasize the propagation-speed inference, physical-layer integration, and scalability. Due to the differences in operating frequency, task dimensionality, and system boundaries, direct comparison across the architectures should be interpreted as indicative rather than absolute. The

optimal architecture choice is inherently application-dependent: the digital electronics for general-purpose computation, the integrated photonics for high-efficiency and fixed-function acceleration, and the metasurface systems for edge-level, wave-based direct processing, and innovative computing form factors. Future progress will hinge on the co-advancing metrics within each paradigm while fostering hybrid systems that combine their respective strengths.

Conclusion

The symbiotic integration of AI and metasurfaces is catalyzing a paradigm shift in both electromagnetic engineering and non-von Neumann computing architectures. This review delineated the bidirectional advancement along two synergistic trajectories: employing AI for the intelligent design and optimization of metasurfaces, and leveraging metasurfaces as the physical platforms for wave-based intelligent computing.

In the first trajectory, deep learning has transcended its role as a mere optimization tool, evolving into a core engine for the inverse design. It has effectively circumvented the computational bottlenecks of the traditional full-wave simulations, enabling the discovery of multifunctional and high-performance meta-atoms and large-scale arrays with unprecedented efficiency. Emerging paradigms like the physics- informed neural networks and generative models are beginning to address the ill-posed nature of inverse designs, offering not just single solutions but rich manifolds of viable structures. The realization of an intelligent and self-adaptive metasurface cloak driven by the deep learning demonstrates [113] the paradigm shift from static design to real-time responsive systems.

The second trajectory marks an even more profound convergence, where the metasurfaces morph from static devices into reconfigurable computational matter. We surveyed the evolution from the free-space diffractive networks to chip-integrated photonic accelerators and programmable microwave metasurface systems. These platforms natively perform the matrix operations and linear transforms at the speed of light, demonstrating orders-of-magnitude advantages in the latency and energy efficiency for specific tasks, as quantified in Sect. 3.4. Notably, this wave-based computing paradigm is transitioning from the proof-of-concept to real-world functionalization. For instance, the programmable metasurfaces are being deployed as the core of intelligent wireless environments for the physical-layer security and adaptive beamforming. Recent work on in-sensor and in-memory wave computing, such as metasurface-enabled direct analog processing of incident scenes, points toward a future where the sensing, communication, and decision-making are fused into a single ultra-efficient physical layer [114].

Despite the remarkable progress, some critical challenges whose solutions will define the next phase of development still persist. For the AI-driven metasurface design, a principal bottleneck remains the reliance on large and high-fidelity datasets and the "black-box" nature of models, which hinders the physical insight and trust. Future efforts must prioritize developing the shared benchmarks and, more importantly, deeper integration

of fundamental physical constraints directly into the learning architectures to foster interpretable and physically-realizable generative models. In parallel, for the metasurface-based computing, the central quest is to achieve general-purpose programmability alongside the energy-efficient nonlinear activation at the hardware level. While the microwave systems advance with active components, the optical systems should seek material-based solutions. Furthermore, scaling systems like PAIM to deeper networks while managing error propagation is non-trivial; this necessitates hierarchical and hybrid designs that strategically combine the analog wave computing with the minimal digital control and explore novel nonlinear optical materials.

Looking forward, the convergence of AI and metasurfaces is poised to yield transformative and mutual insights. On one hand, the metasurfaces provide a unique physical testbed for novel AI concepts, inspiring algorithms through the exploration of extreme spatio-temporal parallelism, the hardware embodiment of neural weights in nanostructures, and physics-based learning in electromagnetic fields. On the other hand, AI is fundamentally redefining the wave engineering by shifting the design philosophy from "what can we analyze?" to "what functionality is demanded?", leading to the discovery of complex and non-intuitive structures and the revolutionary concept of a trainable "meta-agent". This bidirectional synergy promises to unlock the full potential of intelligent electromagnetic matter for the next-generation technologies.

Ultimately, the co-evolution of intelligent design and intelligent computing is closing the loop: better AI designs enable more powerful computational metasurfaces, which in turn can accelerate the next generation of AI hardware. This virtuous cycle promises to unlock the full potential of the intelligent electromagnetic matter, paving a way for disruptive technologies in the ultralow-power edge AI, real-time ubiquitous sensing, adaptive wireless communications, and beyond.

Authors' contributions

Qian Ma and Zirui Feng contributed equally to this work. Jianwei You and Tiejun Cui supervised the project, conceived the review's scope, and secured funding. Junming Hou, Che Liu, Yisu Huang, and Yi Zhang were primarily responsible for the literature review, analysis, and writing of the Intelligent Design section. Qian Ma, Zirui Feng, Yuming Ning, Ruisi Li, Qianwen Wu, and Ze Gu were primarily responsible for the literature review, analysis, and writing of the Intelligent Computing section. The initial draft of the manuscript was prepared by Qian Ma and Zirui Feng, and was critically reviewed, edited, and approved by all authors.

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Data availability

All data are fully available without restriction.

Declarations

Competing interests

The authors declare that they have no competing interests.

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